

Mohamed Irfan M Y

Software Developer — IoT Application Developer — IT Specialist
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LinkedIn — Portfolio — GitHub

Career Summary

Electronics and Communication Engineering student with hands-on experience in software development, IoT systems, embedded firmware, cybersecurity analysis, cloud integration, and real-time monitoring applications. Skilled in Python, JavaScript, Flutter, Firebase, REST APIs, Microsoft Azure fundamentals, embedded systems, networking fundamentals, troubleshooting, and application development. Experienced in developing web and mobile applications, integrating cloud services, performing security analysis, and supporting hardware-software systems.

Education

Kalasalingam University

B.Tech in Electronics and Communication Engineering

2022 – 2026

CGPA: 7.39 / 10

Relevant Coursework: Computer Networks, Operating Systems, Embedded Systems, Internet of Things, Data Communication Networks, Microprocessors and Microcontrollers

Technical Skills

- **Programming:** Python, JavaScript, C, Embedded C
- **Software Development:** OOP, REST APIs, SDLC, Debugging, Problem Solving
- **Frontend & App Dev:** HTML, CSS, JavaScript, Flutter
- **Backend / Cloud:** Flask, REST API Development, Firebase (Auth, Firestore, Database, Storage), Microsoft Azure Fundamentals
- **Database:** Firebase Firestore, Realtime Database
- **IoT & Embedded:** ESP32, STM32, MPU6050, UART, SPI, I2C, ADC, GPIO, Real-Time Systems
- **Cybersecurity:** Android Static Analysis, APK Reverse Engineering, Vulnerability Detection, Threat Analysis
- **Networking:** TCP/IP Fundamentals, DNS, DHCP, HTTP/HTTPS, Network Troubleshooting, Network Support and Security
- **AI & Productivity:** AI Tools, Prompt Engineering, Google AI Essentials, Responsible AI Usage
- **Operating Systems:** Windows, macOS, Linux (Ubuntu, Kali Linux), Android, iOS
- **Tools:** Git, GitHub, VS Code, Android Studio, Firebase, Cisco Packet Tracer, Arduino IDE, STM32CubeIDE

Experience

Project Intern – Microcontroller Firmware & PCB Design

C-DAC – Bengaluru, India

- Developed embedded firmware in C for sensor data acquisition, processing, and communication.
- Implemented UART, SPI, I2C, ADC and GPIO interfaces for real-time embedded applications.
- Performed firmware debugging, hardware validation, and system troubleshooting.
- Supported hardware-software integration, testing, and performance optimization.
- Assisted in PCB bring-up, schematic review, and prototype validation workflows.

Projects

HEAL Lite – Tremor Monitoring System with AI and Hybrid Storage

Dec 2025

IEEE Publication

- Developed a wearable IoT-based tremor monitoring system for real-time health tracking.
- Built a Flutter mobile application for live monitoring and historical analysis.
- Integrated Firebase Authentication, cloud synchronization, and data storage services.
- Implemented ESP32 and MPU6050 sensor integration for real-time data acquisition.
- Designed offline logging support and AI-assisted tremor trend visualization.
- Published research work based on the project in IEEE Xplore.

APKShield – Android Security Analysis Platform

Apr 2026

GitHub Repository

- Developed a Flask-based web platform for Android APK security analysis and vulnerability assessment.
- Extracted AndroidManifest.xml and analyzed 34+ dangerous permissions.
- Implemented 20+ regex-based detection rules for API keys, tokens, and sensitive information.
- Designed a 4-layer binary XML decoding engine for APK inspection.
- Built an interactive dashboard with automated threat scoring and security reporting.
- Generated vulnerability reports to support application security reviews.

Achievements

First Prize – Electrathon (VINTRA'26)

Oct 2026

- Developed a smart home automation system with fire alert detection and remote appliance control through a mobile application.

Runner-Up – IoTropolis (VINTRA'24)

Oct 2024

- Developed an IoT-based gas leakage detection and alert system with cloud notification capabilities.

Additional Skills

Application Support, Technical Support, Troubleshooting, System Integration, Incident Analysis, Root Cause Analysis, IT Operations, Cloud Applications, Software Development, Mobile Applications, Web Applications, Networking Fundamentals, Cybersecurity, Embedded Systems, Firmware Development, Linux, Windows, REST APIs, Version Control, Agile Environment.

Certifications

• Human Resources: Payroll – LinkedIn Learning	2025
• Wadhvani Ignite India – WF Global Certificate	2025
• Fortinet Certified Associate (FCA) – Fortinet ID: 8816711595MI Verify	2026
• Networking Basics – Cisco Networking Academy Credential	2026
• One Million Prompters – Dubai Future Foundation Certificate	2026
• Cybersecurity Bug Bounty Mastery – Infoziant Certificate Verification	2026
• Google AI Essentials – Google Verify	2026
• Google Prompting Essentials – Google Verify	2026
• Network Support and Security – Cisco Networking Academy Credential	2026